VINNOVATEIT ROUND 1 RECRUITMENT CHALLENGE · UI/UX SUBDOMAIN

STILL — The Camera That Waits For You

”Where did my button go? Rethink the Interaction”

A mobile camera paradigm eliminating the physical shutter button. Rather than asking users to execute an anxiety-inducing, tremor-inducing tap command, STILL infers intent from continuous sensor stillness and provides a post-hoc circular buffer rewind.

HOLD-TO-CAPTURE

0.8s (Tunable)

Range: 400–1500 ms

STEADINESS TRIGGER

Motion Heuristic

Simulated in prototype

REWIND BUFFER

3.0s (Simulated)

7 illustrative frames

ACCESSIBILITY

Target & Contrast

44px targets & 18:1 contrast

DELIVERABLE D1

The Removed Interaction & The Problem

OFFICIAL PROBLEM STATEMENT (148 WORDS · LIMIT ≤ 150) Word count verified

Removed: The **shutter button** in the mobile camera app.

Why it existed: The shutter historically fulfilled five critical jobs: declaring unambiguous capture intent, freezing an exact split-second, confirming capture through feedback, offering a fixed thumb target, and maintaining a familiar mental model. Crucially, the physical act of tapping a screen button induces device tremor — causing camera-shake blur at the precise instant of exposure.

The core problem: Without a shutter button, users lack a direct command to trigger capture, risking unintended shots or missed memories. Naive solutions — like volume-key triggers or full-screen tapping — simply relocate the physical button and preserve the shake penalty.

The design question: *How can a camera infer deliberate intent and capture the optimal moment without requiring a destructive command? The tap only sets the focus point; capture waits until the phone is steady.*

The 5 Jobs of the Shutter & How STILL Replaces Them

Shutter Job	Traditional Mechanism	Inherent Flaw / Limitation	STILL's Solution
1. Intent	Explicit tap button: "Take photo now".	High cognitive friction; commits before verifying framing quality.	**Tap-to-Focus + Physical Stillness**: User taps subject; steady hand confirms active intent.
2. Timing	Single discrete shutter click freezes millisecond.	Human reaction time lags behind action, causing missed smiles or blinks.	**Moment Strip (3s Buffer)**: 7 candidate frames recorded pre- and post-lock. Pick your preferred frame post-hoc.

Shutter Job	Traditional Mechanism	Inherent Flaw / Limitation	STILL's Solution
3. Feedback	Screen flash / mechanical shutter click.	Confirms shutter tripped, but gives zero signal on focus sharpness or motion blur.	**Linear Ring Fill + Synthetic Haptic**: Physical progress gauge, crisp audio tick, and subtle screen bloom.
4. Reachability	Fixed circular button at bottom centre.	Requires awkward one-handed thumb stretching or two-handed gripping on tall screens.	**Entire Viewfinder Active**: Anywhere tapped becomes the capture lock point; effortless from any grip.
5. Stability	Physical tap impact on glass or chassis.	**Self-defeating:** Tap momentum directly induces angular jerk and motion blur at exposure.	**Stillness Triggered**: Photo captures automatically after tap jolt has settled and device is steady.

DELIVERABLE D2

Interaction Model & Finite State Machine

STILL operates on a deterministic 6-state finite state machine with automatic recovery paths, ensuring zero accidental shutter triggers and effortless frame retrieval.

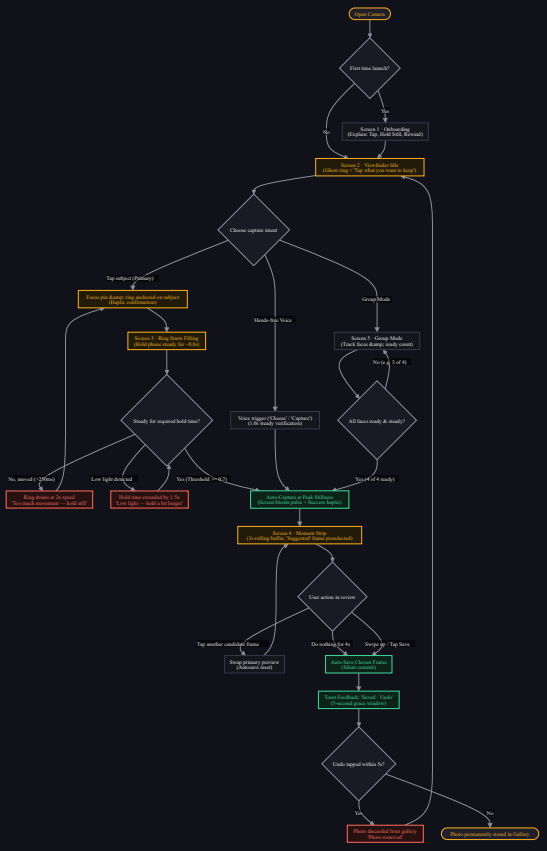


Figure 2.1 — Deterministic State Machine Architecture (Exported from `docs/D2-user-flow.mmd`)

1. IDLE ➔ FOCUSED

Viewfinder displays ghost ring. User taps subject: focus pin anchors at (x,y), light haptic tick fires, 3s timeout guard initializes.

2. LOCKING ➔ CAPTURED

Hand steady: 104px ring fills linearly for 0.8s. If shake > 250ms, ring drains 2× with low buzz. At 100%, trigger fires with shutter chirp.

3. REVIEW ➔ SAVED

Moment Strip opens with preselected Suggested frame (illustrative). 4s countdown auto-saves. 5s toast offers one-tap Undo to discard.

DELIVERABLE D3

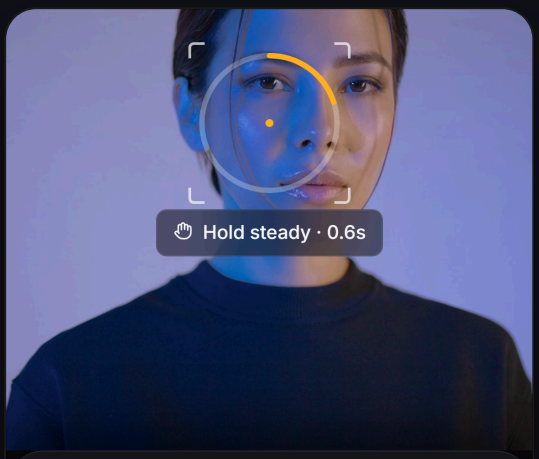
The 5 High-Fidelity Screens Walkthrough

Every screen adheres to STILL's dark-room visual grammar: OLED deep blacks (`#0B0B0F`), signal amber (`#FFB020`), frosted glass chips, and 44×44px minimum interactive touch targets.

SCREEN 01

Progressive Onboarding

Teaching the invisible gesture: First-time launch explains the absence of the button via a 3-step progressive coach card (1. Tap subject, 2. Hold still 0.8s, 3. Rewind moments).



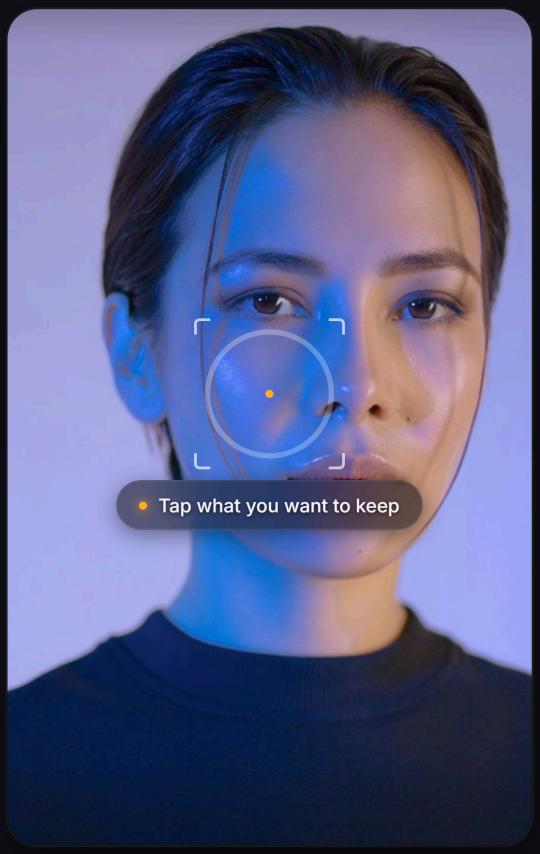
No shutter. Just stillness.

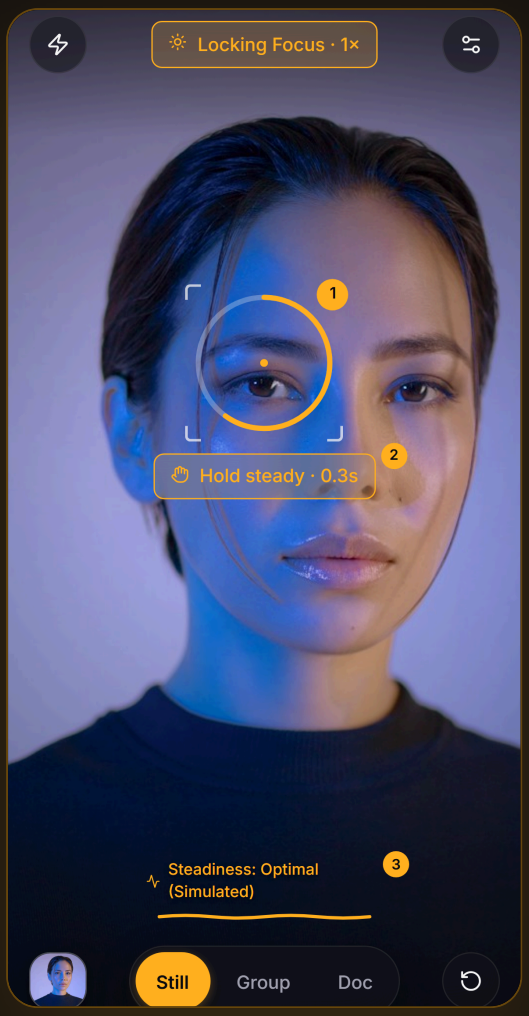
1. **Tap** what matters to set focus.
2. **Hold still** while the ring fills (~0.8s).
3. **Rewind** to pick your best moment afterwards.

SCREEN 02

Zero-Button Viewfinder

The calm canvas: Edge-to-edge optical preview with no distracting white shutter circle. A subtle coachmark "Tap subject to lock" fades smoothly after 1.5s idle dwell.





SCREEN 03 · THE HERO SCREEN

Continuous Stillness Locking (Mid-Action)

This screen demonstrates the heart of STILL's interaction invention. The user has tapped the portrait subject and is holding steady at 60% progress.

1. **104px Physical LockRing & Tracking Brackets**
   Linear sweep (not eased) reflects real-time device steadiness. Dimmed vignette emphasizes the subject.
2. **Dynamic Countdown Glass Chip**
   "Hold steady · 0.5s" microcopy grounds timing expectations, preventing premature movement.
3. **Steadiness Waveform Meter**
   Replaces the old shutter button position. Dynamic sine waveform straightens into a solid bar as device steadiness is reached.

SCREEN 04

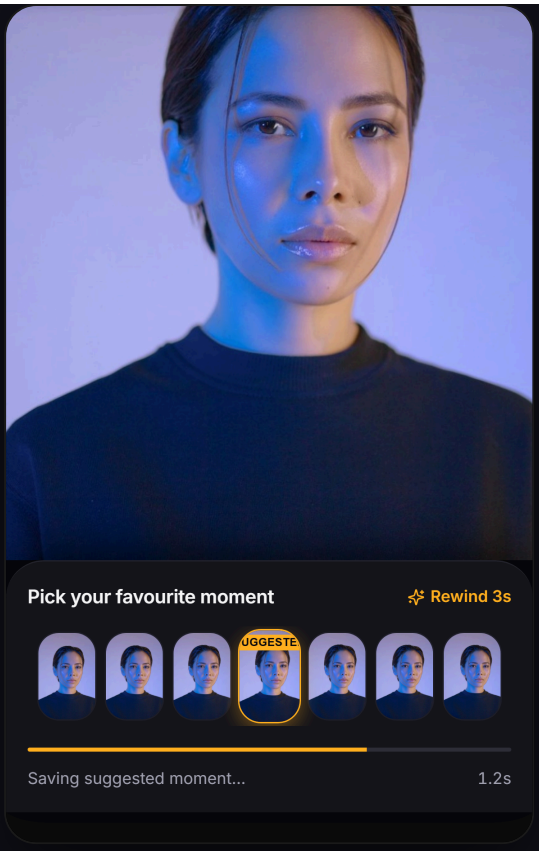
Moment Strip (Rewind & Choose)

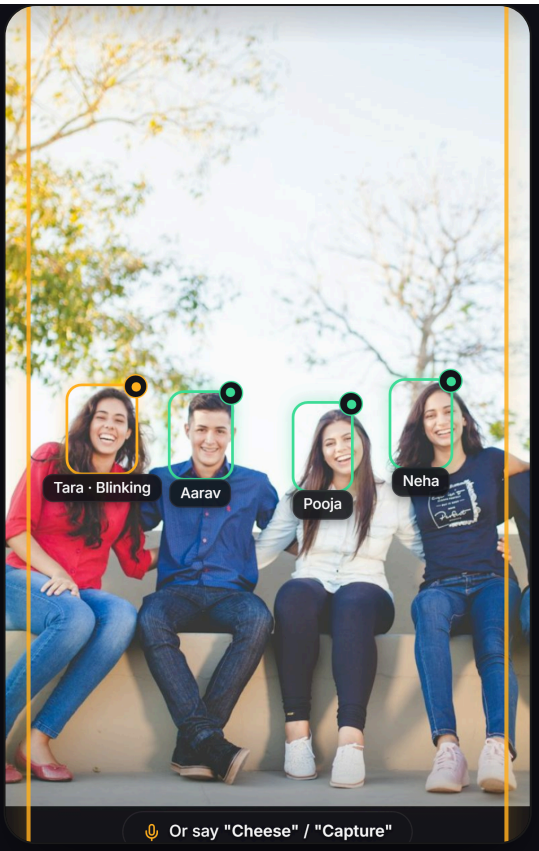
Post-hoc reassurance: 7-frame simulated rewind carousel with preselected Suggested frame (illustrative). 4-second autosave countdown removes decision paralysis; 5-second Undo toast guarantees reversible control.

SCREEN 05

Group Mode (”Everyone’s In”)

Context-aware collective intent: Real-time face tracking for 4 friends with readiness status (green = ready, amber = blinking/eyes closed). Global perimeter ring locks and triggers only when all subjects are ready.


Pick your favourite moment
Rewind 3s
UGGESTE
Saving suggested moment...
1.2s


Tara · Blinking
Aarav
Pooja
Neha
Or say "Cheese" / "Capture"

DELIVERABLE D4

Key Design Decisions

Official submission copy (190 words · Limit ≤ 200 words):

1. **Reuse tap-to-focus.** Everyone already taps to focus, so STILL extends a familiar gesture instead of teaching a new one.
2. **Stillness is consent.** The tap sets the focus point; the photo is taken only after the phone has settled for about 0.8 s. The jolt from the tap fades before capture, and an accidental touch never takes a photo.
3. **Visible, physical progress.** A filling ring shows what is about to happen. If you move, it drains, so you stay in control without a button.
4. **Choose the moment afterwards.** A rewind strip (a simulated 3-second buffer in this prototype) removes the pressure of perfect timing. A suggested frame is preselected, saves after 4 seconds, and Undo lasts 5 seconds.
5. **Whole-screen control.** No corner to reach, which suits one-handed and selfie use.
6. **Alternate input triggers.** Adjustable hold time, simulated voice and key triggers for capture without touching the screen.
7. **Calm dark UI.** The photo is the hero; amber is reserved for action and suggestions.

Note: the timings (0.8 s, 4 s, 5 s) are starting values to tune through usability testing.

INCLUSIVITY & QA

Accessibility & Ergonomics Verification

≥ 44 × 44 px

Touch Target Enforced

All icon buttons, chips, and strip thumbs audited for Apple HIG guidelines (≥ 44×44 px).

18.0 : 1 Contrast

High Contrast Palette

Primary text on dark canvas evaluates to 18.0:1; signal amber to 10.7:1, exceeding standard readability benchmarks.

aria-live="polite"

Screen Reader Region

Announces camera state transitions, countdown seconds, and save confirmations in real time.

Reduced Motion

Stepped Discrete Fill

Replaces continuous ring sweep with 33% → 66% → 100% ticks for vestibular sensitivity.

Competitive Feature Comparison

Capability	Standard Mobile Camera	Burst Mode	Apple Live Photos	STILL (Proposed)
Trigger Method	Physical button tap on screen	Hold shutter button	Button tap + auto video	**Hold still (0.8s intent)**
Tap Camera Shake	High (tapping screen induces camera shake)	High (pressure on glass)	Present on initial tap	**Minimal (capture occurs after phone settles)**
Capture Timing	Lagged behind human reaction	Generates many duplicate photos	3s video slice	**Simulated rewind buffer (3s concept)**
Triage & Cleanup	Manual re-take	Painful manual curation	Key photo scrubber hidden	**Moment Strip with preselected Suggested frame**
Group Coordination	Shout "1, 2, 3 smile!"	Burst & hope someone didn't blink	Static face detection	**Context-aware readiness ring**
Motor Inclusivity	Requires precise finger hit	Requires sustained thumb force	Standard button	**Tap anywhere + Alternate input triggers**

TECHNICAL INTEGRITY

Prototype Limitations (Simulated vs Real Hardware)

STILL is a frontend browser UI/UX prototype created for the VinnovateIT design challenge. To maintain absolute technical truthfulness, prototype behaviors are clearly distinguished from native hardware implementations:

Device Motion Heuristic

Steadiness is evaluated via browser DeviceMotionEvent acceleration or on-screen controls, not a native hardware multi-axis gyroscope.

Rewind Buffer Concept

Moment Strip presents 7 illustrative variations of a high-resolution portrait to demonstrate the 3-second rewind UI/UX workflow, not a hardware circular ring buffer.

Group Readiness Tracking

Face detection bounding boxes and readiness are simulated via interactive state machine models rather than on-device machine vision.

Alternate Input Triggers

Simulated voice command and keyboard triggers still require device steadiness to settle screen tap shake and avoid accidental exposure.
